Nico
Rafael

Emergency Department

Wait-time analysis

A healthcare analytics investigation into ER wait times, using synthetic datasets modeled after Las Vegas hospitals.

Problem & Goal

Leadership at the Valley Regional Medical Center has requested to **investigate and present findings regarding extensive wait-times in their Emergency Department.**

The goal is to **reveal the average ER wait-times, busy hours of the day/week, and relations between patient severity levels and wait-times.**

Additionally, I simulated follow up questions/ answers from the stakeholders.

Tools Used

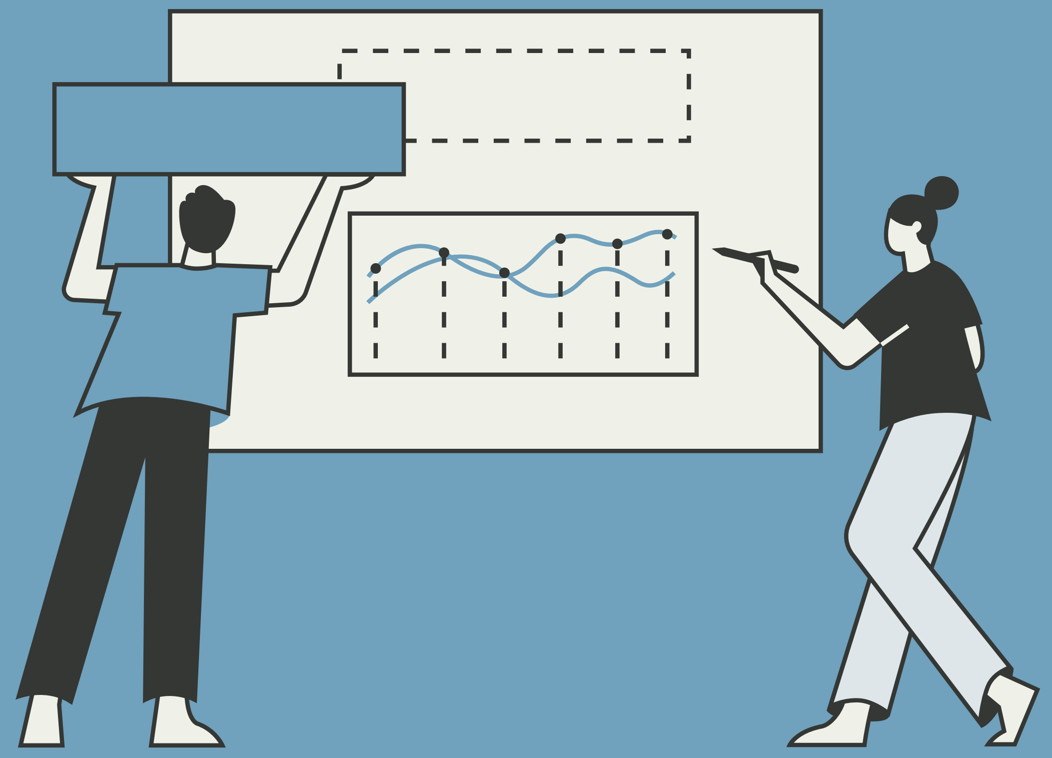
PostgreSQL

Power BI

GitHub

SQL & Modeling Workflow

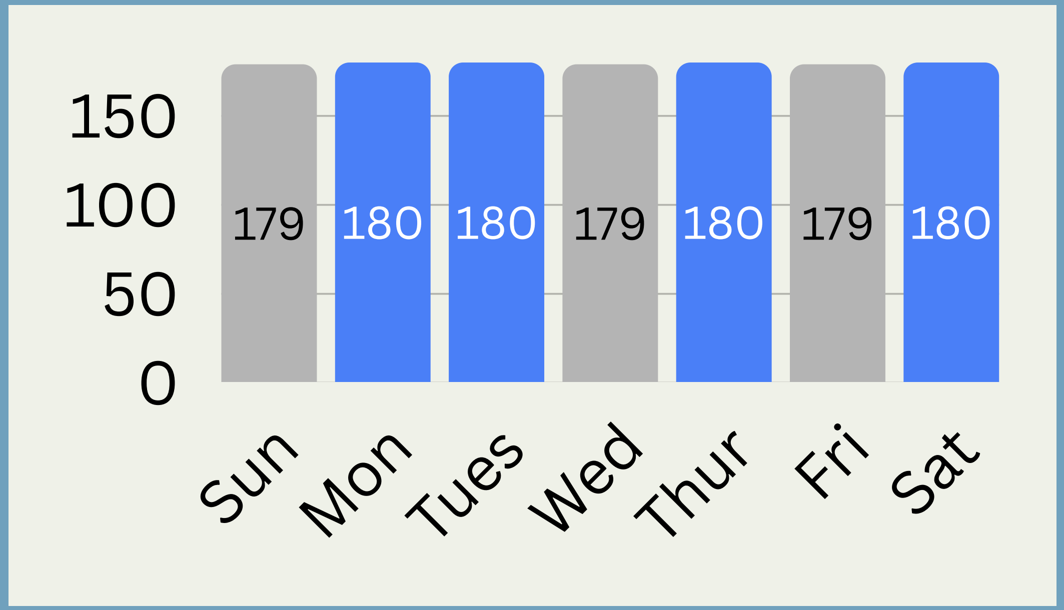
- Designed Entity Relationship Diagram
- Created tables in SQL
- Imported CSVs: ER visits, appointments, patients, and providers
- Standardized providers names within dataset
- Conduct exploratory SQL analysis
- Exported aggregated tables for visualization

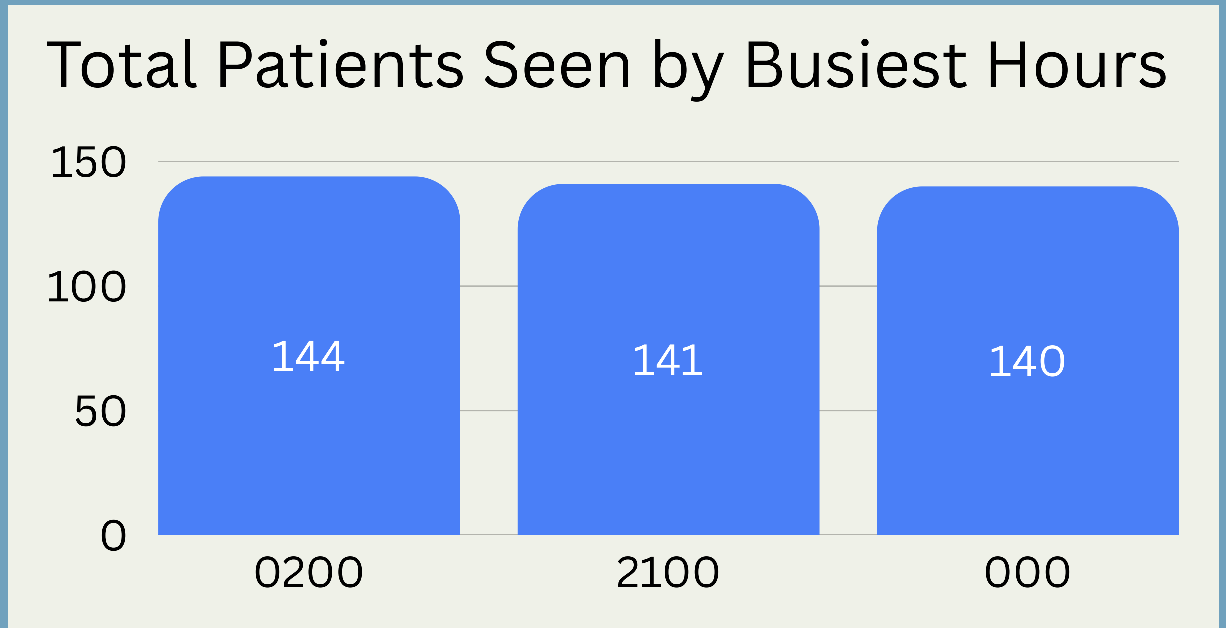


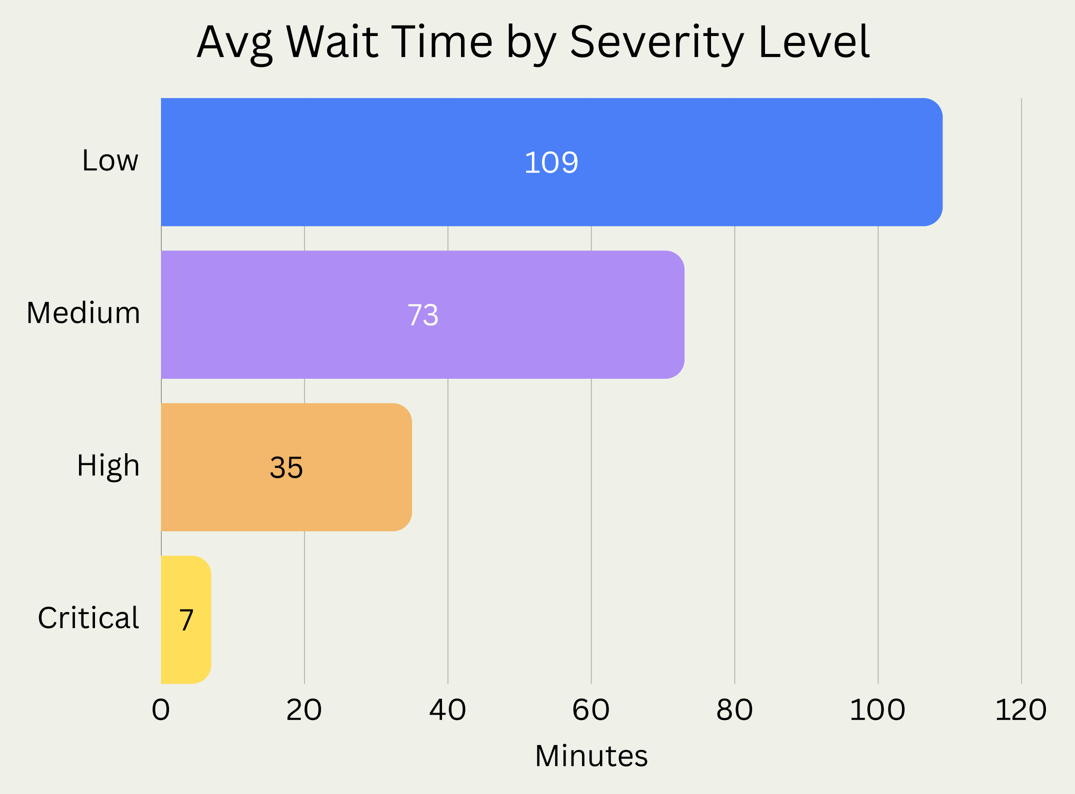
Insights

74 min
Avg Wait

67 min
Median Wait







Longest Wait-Time Experienced

180 min on Mondays, Tuesdays, Thursdays, and Saturdays

71% of Visits During Peak Periods

Generated by Medium- and Low-severity patients

Busiest Hours

12 AM

2 AM

9 PM

Business Recommendations

Ensuring High-Acuity Focus While Optimizing Lower-Acuity Flow

Evaluate staffing and patient flow during peak demand periods: Monday, Tuesday, Thursday, and Saturday, at 12 AM, 2 AM, and 9 PM.

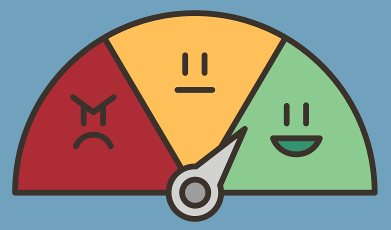
Continue prioritizing Critical- and High-severity patients through the current triage process. This is working good. Monitor Low and Medium-severity patient flow to identify opportunities for operational improvements.

Impact

This analysis equipped stakeholders with clear operational signals behind prolonged wait times, enabling them to pursue targeted follow-up questions and make more informed, data-driven decisions about staffing, peak-period capacity, and patient flow.

Follow Up Insights

After initial findings, stakeholders requested additional insight to determine causes of long wait-times in the emergency department



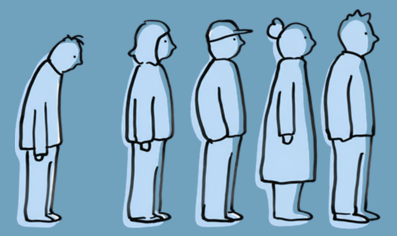
Patient Satisfaction

- Scores reamained stable across appointment types

No-Shows

24%

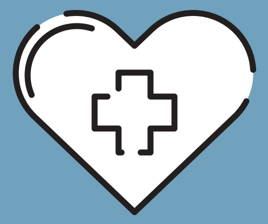
- Older Adult Male/Females from Henderson
- Older Adult Females from Las Vegas



ER Volume

37%

- **Henderson residents** represent the largest share of visits to ER



Wait - Times

45-64

- Patients in these **age ranges** experience the longest waiting

Further investigation is needed to identify the operational drivers behind these trends

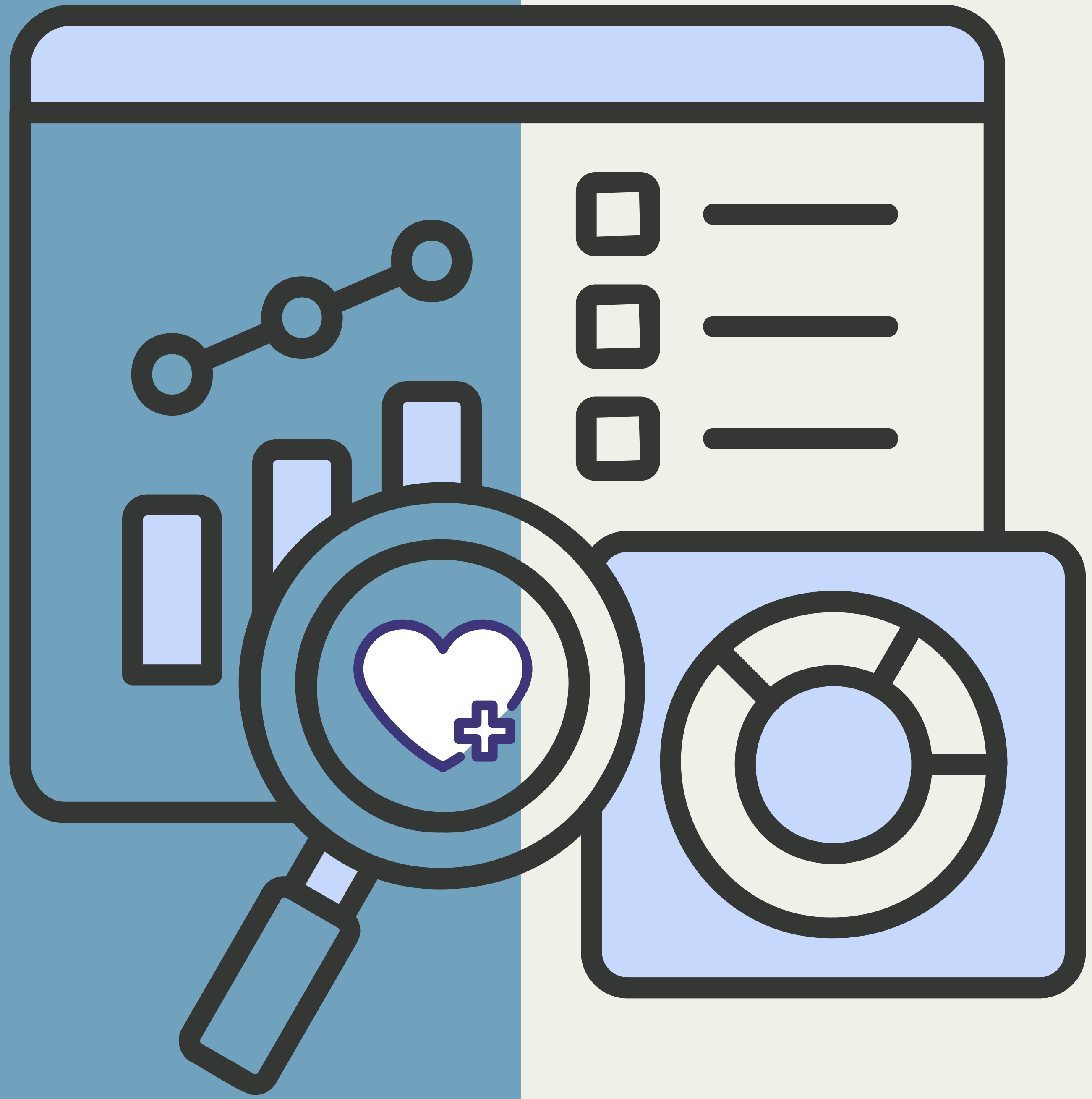
Turning Insights Into Action

While deeper investigation will refine the picture, these findings already give stakeholders clear direction for taking action on peak-period capacity, staffing alignment, and patient flow.

Confirm operational bottlenecks through workflow analysis
Model peak-period capactiy and resoruce needs
Evaluate diagnositc and arrival patterns contributing to delays
Map patient journey through the ER to identify friction points

Thank You

Let’s build smarter decisions together



[GitHub]
[LinkedIn]
nrafael@gmail.com